Venkata Lakshmi Parimala Pasupuleti

Applied AI Solutions Architect · Multi-Agent Orchestration · MCP · A2A · RAG · Prompt Engineering · Amazon Bedrock · LangGraph · AWS · Python

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Virginia, USA — Open to Travel 25-40%

PROFESSIONAL SUMMARY

Applied AI Solutions Architect with 8+ years of production engineering and 3+ years hands-on agentic AI implementation — building multi-agent orchestration systems, RAG pipelines, MCP-compatible tool integrations, and A2A (Agent-to-Agent) communication patterns using LangGraph, FastAPI, and foundation models (Anthropic, OpenAI, Groq, Ollama). Built **AgentProbe**: a 3-agent LangGraph runtime with Orchestrator → RAG Tracer → Auto Fix inter-agent handoffs, tool calling, function calling, and multi-provider LLM routing — directly mirrors Amazon Connect's agentic AI architecture. Built **Leasel**: production RAG knowledge base system achieving **93.6% precision** across 10 compliance rules on 1,240 documents with LLM-as-judge model evaluation. Deployed **Amazon Bedrock-compatible** multi-provider routing (Anthropic/OpenAI API). Expert in data readiness, knowledge base configuration, guardrail design, and prompt engineering. AWS production infrastructure (S3, EC2, RDS, Glue) 7+ years; **200+ SQL optimizations** across 1M+ records. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer — Agentic AI Systems · MCP-Compatible Tool Integration · RAG Knowledge Base · A2A Workflows

- Built an agentic mission-planning system as a **LangGraph stateful graph** (parse → plan → retrieve → validate → conditional routing) — **MCP-compatible tool integration** design exposing APIs and data sources in a standardized format for dynamic agent tool discovery; implemented **A2A communication patterns** for multi-agent workflow routing between planning, retrieval, and validation agents.
- Implemented a **RAG knowledge base** (retrieval over FAA Part 107 corpus) grounding every compliance decision in cited source documents; applied **function calling** and **guardrail** validation enforcing operational constraints before execution; conducted **data readiness** assessment ensuring data assets were structured for reliable AI agent retrieval.

George Mason University
Aug 2024 – Aug 2025

Research Associate — AI/ML Modeling · Model Evaluation · Stakeholder Reporting

- Built quantitative prediction models (Python, R) achieving **$R^2 = 0.977$** across 190+ countries; applied **model evaluation** frameworks including cross-validation, hypothesis testing, and performance benchmarking against defined success criteria before production deployment.
- Surfaced anomalies across 3 research pipelines using **Isolation Forest**; designed strategic dashboards (Tableau, ggplot2) improving decision-making comprehension **40%**; mentored **600+ students** in Python, data analytics, and database design.

Entrata
Sep 2018 – Aug 2024

Senior Software Engineer — Production AI Pipelines · AWS Cloud Architecture · Full-Stack · Compliance Automation

- Shipped **production AI pipeline** processing **1.6M financial records** (Python, PyTorch, OpenAI API) — automated data workflows with output validation, quality gates, and **data readiness** controls; shipped 20+ full-stack modules (PHP, Python, PostgreSQL, JavaScript, RabbitMQ) for 100K+ monthly active users at 99.5% uptime.
- Architected AWS cloud infrastructure (S3, EC2, RDS, Glue) with **serverless-compatible** pipeline design; achieved **100% PHI compliance** (Libsodium encryption, zero security incidents); optimized **200+ SQL queries** (30% performance gain); built **third-party integrations** (Salesforce CRM, API connections, data pipelines) and led **microservices architecture** redesign (Docker, service-based architecture — 50% scalability improvement).
- Built internal **CI/CD pipelines** (GitHub Actions); conducted 300+ peer code reviews; mentored 5+ engineers; eliminated **80% of manual operational workload** through Python/Bash automation across 120+ Agile sprints.

PROJECTS

AgentProbe: Multi-Agent AI Testing Toolkit — A2A · MCP-Compatible · LLM-as-Judge

Python · LangGraph · LangChain · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent **LangGraph** runtime (Orchestrator → RAG Tracer → Auto Fix) with **A2A (Agent-to-Agent)** inter-agent state handoffs, **tool calling**, **function calling**, and **multi-provider foundation model** routing (Anthropic, OpenAI, Groq, Ollama) — **MCP-compatible** architecture for standardized tool discovery and invocation; 15-endpoint **serverless-ready FastAPI** backend running at zero cost locally.
- Built 5-metric **LLM-as-judge model evaluation** engine (factual accuracy, hallucination, retrieval, completeness, compliance) with **CI/CD** (GitHub Actions) — every commit evaluated automatically; React dashboard with **ChromaDB RAG knowledge base**, trace-level **agent observability**, retrieval quality, and latency metrics across concurrent agent test suites.

Leasel: Production RAG Compliance System — 93.6% Precision · Knowledge Base · Agentic Pipeline

Python · LangGraph · RAG · ChromaDB · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end agentic RAG pipeline with **knowledge base** configuration (1,240 documents, 5 US states + federal standards) — **prompt engineering**, **guardrail** design, **data readiness** validation, and LLM-as-judge **model evaluation** (pass@1 81.7%, pass@3 88.3%); achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination fully auditable with data lineage.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Machine learning, agentic AI systems, RAG pipelines, model evaluation, cloud infrastructure, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Agentic AI & Architecture: Multi-Agent Orchestration (LangGraph, LangChain), A2A (Agent-to-Agent) Communication, MCP (Model Context Protocol) Tool Integration, RAG Pipelines, Knowledge Base Configuration, Prompt Engineering, Foundation Model Selection (Anthropic, OpenAI, Groq, Ollama), Tool Use & Function Calling, Guardrail Design, Model Evaluation (LLM-as-Judge, pass@k), Agent Observability & Tracing, Chain-of-Thought Reasoning, Data Readiness Assessment

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue) — Amazon Bedrock-compatible; FastAPI (serverless-ready), Docker, CI/CD (GitHub Actions), Microservices, API integrations, PostgreSQL, MySQL, ChromaDB (vector knowledge base), Embeddings, Semantic Search, Python (expert), SQL (expert — 200+ production optimizations), R, JavaScript, PHP, React, Bash, Linux

ML & Analytics: PyTorch, TensorFlow, Scikit-learn, Isolation Forest, Random Forest, XGBoost, BERT, spaCy, Hugging Face, NLP, Feature Engineering, Statistical Modeling; Tableau, Power BI, ggplot2; Agile/Scrum (120+ sprints)